
Supplementary information

A fully open AI foundation model applied to chest radiography

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Supplementary Methods

A Ablation Study

To showcase the advantages of the Ark⁺ framework, we conducted ablation studies that highlight its multi-task head design (Sec. A.1) and cyclic training approach (Sec. A.2). Additionally, to further illustrate the performance improvements achieved with Ark⁺, we provide fine-tuning baselines initialized from an ImageNet-pretrained model for comparison (Sec. A.3).

A.1 Multi-task Heads vs. Single-task Head

Compared with a single-task head, the multi-task head design of Ark⁺ delivers superior performance (see Supplementary Table 2) and provides several key benefits:

1. Reduce manual effort: The multi-task heads design eliminates the need for prior label consolidation: “understanding” and pooling all labels from different datasets into a predefined list (see Supplementary Table 1), simplifying the process.
2. Increase flexibility and scalability: Pluggable heads offer easy integration of new tasks without requiring retraining or extending the head’s dimension.
3. Lower code complexity: Task-specific heads avoid the complexity of managing class indices for loss computation, unlike the single-task head, which must map predictions to specific datasets’ ground truth.

Supplementary Table 1 Manually assembling the labels from five datasets into a predefined list.

Index	Task 1 [CheXpert]	Task 2 [ChestX-ray14]	Task 3 [RSNA Pneumonia]	Task 4 [VinDr-CXR]	Task 5 [Shenzhen]
0	No Finding		Normal	No finding	
1	Enlarged Cardiomediastinum				
2	Cardiomegaly	Cardiomegaly			
3	Lung Opacity		Lung Opacity		
4	Lung Lesion				
5	Edema	Edema			
6	Consolidation	Consolidation			
7	Pneumonia	Pneumonia		Pneumonia	
8	Atelectasis	Atelectasis			
9	Pneumothorax	Pneumothorax			
10	Pleural Effusion	Effusion		Pleural Effusion	
11	Pleural Other				
12	Fracture				
13	Support Devices				
14		Infiltration			
15		Mass			
16		Nodule			
17		Emphysema			
18		Fibrosis			
19		Pleural Thickening			
20		Hernia			
21			No Lung Opacity /Not Normal		
22				Lung tumor	
23				Tuberculosis	Tuberculosis
24				Other diseases	

A.2 Cyclic Training vs. Simultaneous Training

Cyclic training enhances model performance more effectively than simultaneous training. Although all data are centralized and accessible, we chose cyclic training over simultaneous training to enhance model performance across multiple tasks. Simultaneous training involves updating the model using a combined loss from all tasks, which can lead to conflicting gradients during back-propagation. This conflict may weaken the overall gradient signal, causing slower convergence and suboptimal performance [1]. By contrast, cyclic training allows the model to focus on one task at a time in each iteration, reducing gradient interference and promoting more efficient learning. To demonstrate this, we conducted an ablation study comparing cyclic training with simultaneous training. In the simultaneous training scenario, mini-batches are constructed by randomly sampling from all datasets, and losses are calculated based on the respective dataset IDs and labels. As shown in Supplementary Table 3, Ark⁺ with cyclic training demonstrates superior performance over simultaneous training. This confirms that cyclic training is a more effective approach in this context.

Supplementary Table 2 Ablation study comparing the multi-task heads design with the single-task head used in Ark⁺.

<i>Test using the pretrained head(s)</i>		Task 1	Task 2	Task 3	Task 4	Task 5
		[CheXpert]	[ChestX-ray14]	[RSNA Penu.]	[VinDr-CXR]	[Shenzhen]
Pretraining tasks	Design	mAUC (%)	mAUC (%)	ACC (%)	mAUC (%)	AUC (%)
Task 2, 4	Multi-task heads	–	79.89	–	94.14	–
	Single-task head	–	79.58	–	93.45	–
Task 1, 2, 3, 4, 5	Multi-task heads	88.12	81.35	74.05	95.5	99.6
	Single-task head	88.09	80.94	73.57	94.96	98.06
<i>Linear-probing</i>		ChestDR		VinDr-CXR (all 28 classes)		
Pretraining tasks	Design	mAUC (%)		mAUC (%)		
Task 2, 4	Multi-task heads	80.75±0.13		91.25±0.09		
	Single-task head	80.21±0.13		90.92±0.06		
Task 1, 2, 3, 4, 5	Multi-task heads	82.24±0.12		93.28±0.06		
	Single-task head	82.42±0.05		92.82±0.07		

As presented in Supplementary Table 1, labels from different datasets were manually consolidated to train a single-task head for comparison with Ark⁺’s multi-task heads. All other experimental settings remained identical. To reduce training costs, we used the Swin-Base backbone with an input resolution of 224×224. Two sets of experiments were conducted using two and five pretraining tasks. After pretraining, we evaluated the heads on hold-out test data of the pretraining tasks through end-to-end inference and conducted linear-probing on the ChestDR and VinDr-CXR datasets to assess embedding quality. The results show that Ark⁺ with multi-task heads consistently outperforms the single-task head, except in the external evaluation on ChestDR for models pretrained with five tasks.

Supplementary Table 3 Ablation study comparing cyclic training with simultaneous training using five pretraining tasks (as listed in Supplementary Table 1).

<i>Test using the pretrained head(s)</i>		Task 1	Task 2	Task 3	Task 4	Task 5
		[CheXpert]	[ChestX-ray14]	[RSNA Penu.]	[VinDr-CXR]	[Shenzhen]
Pretraining tasks	Method	mAUC (%)	mAUC (%)	ACC (%)	mAUC (%)	AUC (%)
Task 1, 2, 3, 4, 5	Cyclic training	88.12%	81.35%	74.05%	95.50%	99.60%
	Simultaneous training	87.02%	80.93%	71.61%	94.37%	98.62%
<i>Linear-probing</i>		ChestDR		VinDr-CXR (all 28 classes)		
Pretraining tasks	Method	mAUC (%)		mAUC (%)		
Task 1, 2, 3, 4, 5	Cyclic training	82.24±0.12		93.28±0.06		
	Simultaneous training	76.54±0.72		91.47±0.15		

A.3 Performance Boosts via Ark⁺ Pretraining

Ark⁺ is designed to train robust and high-performing foundation models by aggregating numerous datasets, both large and small, to learn from their heterogeneous annotations. We consciously included smaller datasets, such as VinDR-CXR and Shenzhen, to demonstrate that Ark⁺ can significantly enhance performance compared to fine-tuning each task individually, particularly for these smaller datasets. As shown in Supplementary Table 4, when fine-tuning the ImageNet-pretrained model on VinDR-CXR and Shenzhen, the performances were 91.41±0.96% and 96.43±0.81%, respectively. After pretraining with Ark⁺, the performance improved to 96.42±0.10% on VinDR-CXR and 99.07±0.06% on Shenzhen. These results validate our approach, demonstrating that pretraining on multiple datasets, including smaller datasets, substantially mitigates overfitting and improves generalizability, especially for bottleneck datasets.

Supplementary Table 4 Ablation study demonstrating performance improvements across six tasks achieved with Ark⁺.

Task	Metric	Fine-tuning baseline	Ark ⁺	Performance boost
MIMIC-II	mAUC(%)	79.67±0.28	80.67±0.16	+1.00
CheXpert	mAUC(%)	88.08±0.14	89.67±0.09	+1.59
ChestX-ray14	mAUC(%)	82.98±0.16	84.43±0.09	+1.45
RSNA Pneumonia	ACC(%)	74.81±0.40	75.73±0.27	+0.92
VinDr-CXR	mAUC(%)	91.41±0.96	96.42±0.10	+5.01
Shenzhen	AUC(%)	96.43±0.81	99.07±0.06	+2.64

The results compare fine-tuning baselines initialized from an ImageNet-pretrained model against the Ark⁺ model, both using the Swin Transformer Large backbone and an input resolution of 768 × 768. Ark⁺ significantly boosts performance across tasks, particularly for smaller datasets such as VinDr-CXR and Shenzhen.

References

[1] Ma, D., Pang, J., Gotway, M.B., Liang, J.: Foundation ark: Accruing and reusing knowledge for superior and robust performance. In: International Conference on Medical Image Computing and Computer-Assisted Intervention, pp. 651–662 (2023). Springer